

JAYASUDHAN M

AI Ops Engineer | MLOps | Production AI Systems

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Coimbatore, Tamil Nadu, India

PROFESSIONAL SUMMARY

AI Ops Engineer with 2.5+ years of experience building and operating production-grade AI systems. Proven expertise in MLOps, AI pipeline reliability, automation, and cloud-native deployment. Strong focus on clean system design, observability, cost efficiency, and real-world scalability. Experienced in healthcare AI, GenAI security, and high-throughput inference systems.

TECHNICAL SKILLS

AI Ops & MLOps: Model deployment, CI/CD for ML, Experiment Tracking, Drift Awareness, Rollback-safe releases

Cloud & DevOps: Docker, Kubernetes, AWS (EC2, S3), OCI, GitHub Actions, Terraform

Backend & APIs: Python, FastAPI, Flask, REST APIs, Async & Sync Inference Services

AI/ML: Scikit-learn, NLP, RAG, LLM Orchestration, TF-IDF, SVD, Ensemble Models

Data: Pandas, NumPy, Power BI, Plotly, MLflow

Governance: GenAI governance, Runtime policy enforcement, Workflow automation

PROFESSIONAL EXPERIENCE

TS Techy

AI Ops Engineer

Remote

Oct 2023 – Present

- Built and operated production-ready AI pipelines, integrating ML models into business workflows for real-world clients.
- Implemented experiment tracking, evaluation metrics, and deployment readiness checks across ML systems.
- Developed AI-driven automation, anomaly detection systems, and real-time dashboards to monitor model health.
- Supported healthcare-tech and IoT-ML solutions, collaborating with delivery teams to ship client-facing AI.

TwirlTact Technology Solutions

IoT with ML Intern

Coimbatore, India

May 2025 – Jun 2025

- Built real-time IoT systems integrated with Machine Learning models for edge computing scenarios.
- Implemented end-to-end data collection, preprocessing, and predictive modeling pipelines.
- Supported edge and cloud ML workflows using Python and microcontrollers.

TS Techy

Project Manager Intern

Remote

Aug 2023 – Oct 2023

- Coordinated cross-functional AI and engineering teams using Agile practices to ensure timely delivery.
- Managed sprint planning, timelines, and delivery tracking, translating business requirements into technical tasks.

KEY PROJECTS

MLOps System for Clinical Trial Risk Prediction | Python, FastAPI, Docker, AWS

- Developed an MLOps platform to predict patient dropout risk, achieving **85% accuracy** with 3-tier risk stratification.
- Built a FastAPI inference service and Dockerized the solution, reducing image size by **78%** (to 655MB).
- Integrated CI/CD and AWS deployment readiness, driving a potential **40% cost reduction** for pharma trials.

PyGenGuard – GenAI Runtime Security Framework | GenAI Ops, Governance

- Designed a deterministic security layer acting as middleware between applications and LLMs.
- Enforces intent, cost, compliance, and trust policies before and after execution to ensure AI safety.

FlixMood – AI Content Recommendation System | Scikit-learn, Streamlit, SVD

- Engineered a hybrid recommendation engine combining Collaborative Filtering and TF-IDF.
- Achieved **72%+ recommendation accuracy** using time-aware and energy-based logic.

RAG Chatbot using GROQ | LLM, RAG, Semantic Retrieval

- Built a high-speed RAG system leveraging GROQ LPU for ultra-low latency inference.
- Implemented PDF ingestion and semantic retrieval for a seamless conversational interface.

EDUCATION

Sri Krishna College of Technology

B.Tech in Artificial Intelligence & Data Science

Coimbatore, India

2022 – 2026

Indian Institute of Technology, Ropar

Minor in Artificial Intelligence

Hybrid

2025 – 2026

CERTIFICATIONS

Oracle Cloud Infrastructure 2025: DevOps Professional | Generative AI Professional | Data Science Professional

Confluent Certified Developer: Apache Kafka